Yichao Cai

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Summary

Ph.D. candidate at AIML (University of Adelaide, 2023–Present) focusing on vision–language representation learning and contrastive learning. Published at NeurIPS (Spotlight), ECCV, and ICLR. Industry experience (Tellhow, Huawei) building and deploying CV/ML solutions and production-grade software.

Research Focus

- **Vision–language models:** image–text representations for cross-modal retrieval and transfer; evaluation and benchmarking.
- **Multi-view/contrastive learning:** InfoNCE theory and practice; diagnostics of training dynamics and failure modes.
- **Robustness to data imperfections:** Data misalignment, underspecification; practical mitigations.
- **Interpretability:** concept learning and interpretable factors for model understanding.

Selected Publications

- **Yichao Cai**, Yuhang Liu, Erdun Gao, Tianjiao Jiang, Zhen Zhang, Anton van den Hengel, Javen Qinfeng Shi. *On the Value of Cross-Modal Misalignment in Multimodal Representation Learning.* NeurIPS 2025 (Spotlight) (lead author) [OpenReview]
- **Yichao Cai**, Yuhang Liu, Zhen Zhang, Javen Qinfeng Shi. *CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts.* ECCV 2024 (lead author) [Springer]
- Yuhang Liu, Dong Gong, **Yichao Cai**, Erdun Gao, Zhen Zhang, Biwei Huang, Mingming Gong, Anton van den Hengel, Javen Qinfeng Shi. *I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?* ICLR 2026 (co-author) [OpenReview]
- **Yichao Cai**, Zhen Zhang, Yuhang Liu, Javen Qinfeng Shi. *The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence.* Preprint, 2026 (lead author) [arXiv]

Technical Skills

ML/AI: PyTorch; vision–language modeling; contrastive learning; representation learning; deep learning; computer vision pipelines
Programming/Tools: Python (NumPy, HuggingFace); C/C++; Git; Linux; Docker; LaTeX
Research/Engineering: experiment design; evaluation; reproducibility; dataset curation; experiment tracking (W&B)

Experience

Postgraduate Researcher, Australian Institute for Machine Learning (AIML) *Sep 2023–Present*

- Develop methods and theory for robust, interpretable vision–language representations; design controlled benchmarks and diagnostics; run empirical studies with rigorous ablations and reporting.

Independent Study (Pre-PhD) *Sep 2022–Sep 2023*

- Intensive preparation in probability, optimization, causal inference, and deep learning; implemented core ML models and training pipelines.

AI Engineer, Tellhow Software *Jun 2020–Aug 2022*

- Built inspection and safety-analytics solutions for power-industry equipment using computer vision and automated analysis workflows. Delivered end-to-end features from prototyping to deployment in collaboration with engineering stakeholders.

Software Engineer, Huawei Technologies *Jul 2019–Apr 2020*

- Developed and maintained communication service components; implemented production features and system integrations.

Visiting Student Researcher, California PATH, UC Berkeley *May 2018–Sep 2018*

- Conducted research on visual perception for autonomous driving under Dr. Ching-Yao Chan.

Education

Ph.D. in Computer Science, University of Adelaide *Sep 2023–Present*
Advisor: Prof. Javen Qinfeng Shi

M.Sc. in Instrument Science and Technology, Wuhan University of Technology *Sep 2016–Jun 2019*
Advisor: Prof. Xiao Zhou

B.Eng. in Measurement & Control Engineering, Wuhan University of Technology *Sep 2012–Jun 2016*

Honors and Awards

NeurIPS 2025 Scholar Award, Neural Information Processing Foundation *Oct 2025*
University of Adelaide Research Scholarships *Sep 2023*
Award for Outstanding Graduates, Wuhan University of Technology *Jun 2019*

Additional

Academic Service: Reviewer for TMLR, ICLR 2026, ICML 2026
Teaching: Guest Lecturer & Head Tutor (Statistical Machine Learning, 2025); Teaching Assistant (2025-2026)